

L-shape routes

RUDY paints the whole bbox

The idea

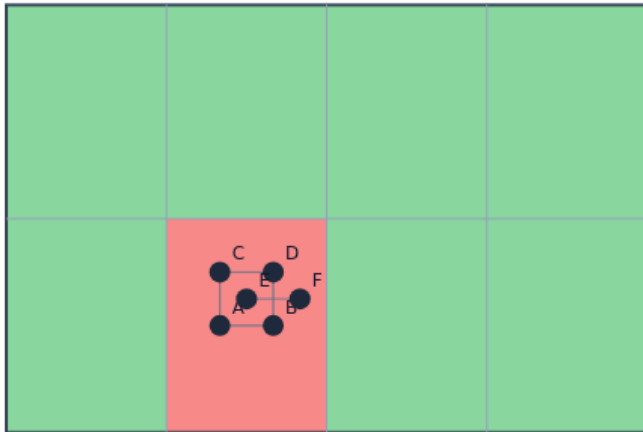
- For a two-pin net
- Multi-pin nets: star from the bbox center to each pin and deposit like two-pin edges
- Compare the resulting matrix to RUDY on the same placement

L-shape idea

PROBABILISTIC DEMAND

L-shape idea

Step 1 / 5 · lshape · module02-03-probabilistic-demand



Explanation

Two-pin nets route on L-shapes with half probability each bend.

- H-then-V and V-then-H

Corridors not filled boxes

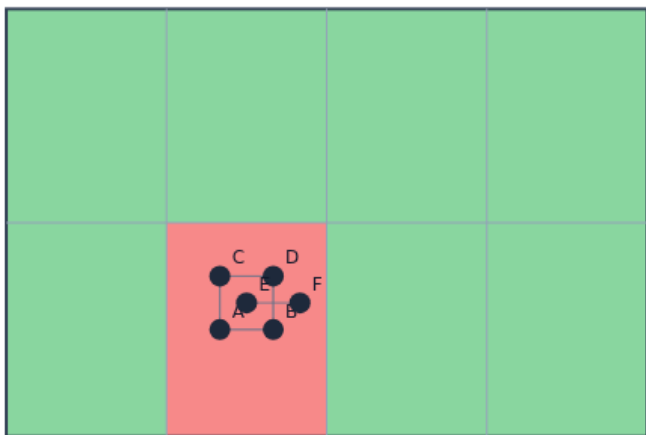
L-shape idea

Deposit along legs

PROBABILISTIC DEMAND

Deposit along legs

Step 2 / 5 · deposit · module02-03-probabilistic-demand



Explanation

Walk GCells on each leg and share demand along the path.

- Corner may be shared

Document scaling

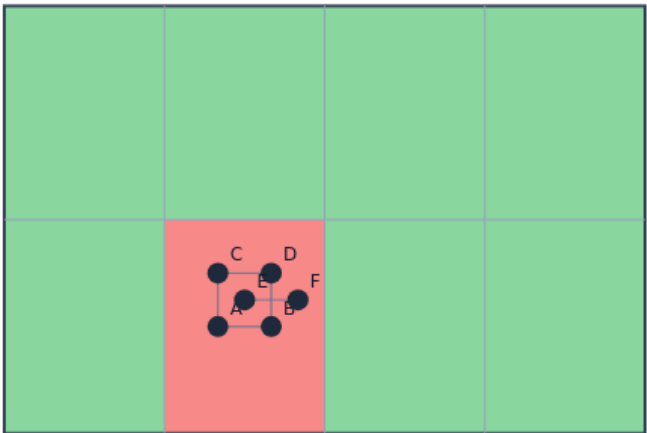
Deposit along legs

Multi-pin star

PROBABILISTIC DEMAND

Multi-pin star

Step 3 / 5 · multi · module02-03-probabilistic-demand



Explanation

Star from bbox center to each pin; deposit like two-pin edges.

- Toy multi-pin

Same Cap

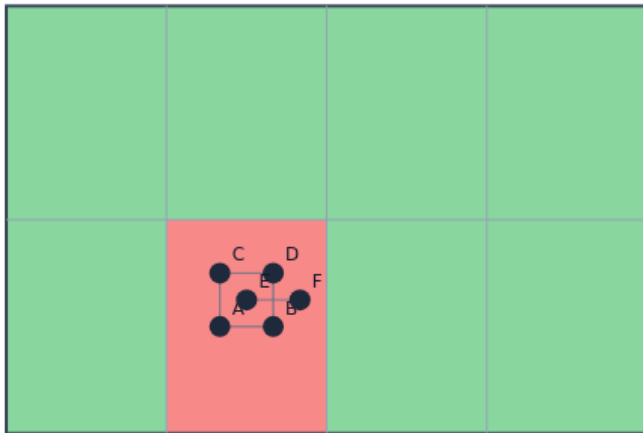
Multi-pin star

Versus RUDY

PROBABILISTIC DEMAND

Versus RUDY

Step 4 / 5 · compare · module02-03-probabilistic-demand



Explanation

Probabilistic concentrates on corridors; RUDY paints the bbox.

- Totals may differ

Both teach overflow

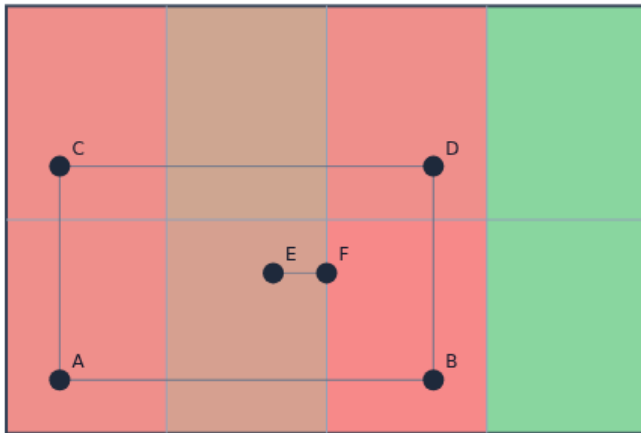
Versus RUDY

Spread again

PROBABILISTIC DEMAND

Spread again

Step 5 / 5 · cool · module02-03-probabilistic-demand



Explanation

Spreading still reduces probabilistic overflow—feedback is placement.

- Same push ideas

Next: heat map

Spread again

Browser lab track

EDA Algorithms Platform

HomeTools

Home / Tools / Probabilistic routing demand

INTERACTIVE TOOL

Probabilistic routing demand

L-shape probabilistic demand versus RUDY on the same placement.

Your job: this lab scores *probabilistic* L-shape demand. Compare mentally to RUDY (metrics show both).

Reset workspace

Challenges0 / 8 cleared

Pick one [Intro] Prob demand > 0

Pick a challenge and click **Start**. Move cells, then **Check**. Reveal golden is study-only.

Start

Show hint

Check

Next challenge

Stop checking

Idle

A B C D E F ← → ↑ ↓

Reset to starter

Reveal golden (study)

Run feedback

Graph

Reference / metrics

selected: A @ GCell (1,0)
HPWL: 7.0
overflow total: 10.00 · max: 10.00 · count: 1
capacity: 2
view: your placement
RUDY overflow total: 5.00 (compare)

Browser lab starter

learn_congestion — 03 probabilistic demand

Implement track

- Implement ``probabilistic_demand``
- On spread placement, print both RUDY and probabilistic totals
- Explain one tile where they disagree and why

Pitfalls

- Double-counting the corner GCell on both L legs
- Forgetting multi-pin nets
- Document your unit choice

Your turn

- Clear the checklist
- Next: turn demand into a congestion heat map